

SET 6

1. R: Create two groups of exam scores; calculate group-wise mean, median, range and SD and determine consistency.

The screenshot shows the RStudio interface with the following R code in the console:

```
# Create two groups of exam scores
> Group_A <- c(78, 85, 90, 72, 88, 95, 80, 84, 91, 87)
> Group_B <- c(65, 70, 75, 68, 72, 78, 74, 69, 71, 76)

# Group A statistics
> mean_A <- mean(Group_A)
> median_A <- median(Group_A)
> range_A <- diff(range(Group_A))
> sd_A <- sd(Group_A)

# Group B statistics
> mean_B <- mean(Group_B)
> median_B <- median(Group_B)
> range_B <- diff(range(Group_B))
> sd_B <- sd(Group_B)

# Display results
> cat("Group A:\n")
Group A:
> cat("Mean =", mean_A, "\n")
Mean = 85
> cat("Median =", median_A, "\n")
Median = 86
> cat("Range =", range_A, "\n")
Range = 23
> cat("SD =", sd_A, "\n\n")
SD = 6.815016

> cat("Group B:\n")
Group B:
> cat("Mean =", mean_B, "\n")
Mean = 71.8
> cat("Median =", median_B, "\n")
Median = 71.5
> cat("Range =", range_B, "\n")
Range = 13
> cat("SD =", sd_B, "\n\n")
SD = 3.994441

# Determine consistency
> if(sd_A < sd_B) {
+   cat("Group A is more consistent.\n")
+ } else {
+   cat("Group B is more consistent.\n")
+ }
Group B is more consistent.
```

The Environment pane on the right shows the following variables:

Variable	Class	Value
Group_A	num [1:10]	78 85 90 72 88 95 80 84 91 87
Group_B	num [1:10]	65 70 75 68 72 78 74 69 71 76
j	num	3
MAD	num	6.16326530612245
mean_A	num	85
mean_B	num	71.8
mean_x	num	24.1428571428571
median_A	num	86
median_B	num	71.5
normalized	num [1:5]	0.125 0.25 0.375 0.5 0.75
range_A	num	23
range_B	num	13
sd_A	num	6.81501610008696
sd_B	num	3.99444058105206
x	num [1:7]	14 18 20 23 27 31 36

2. R: Construct a Q-Q plot for 21,24,27,29,31,34,37,41,45,48 and comment on normality.

The screenshot shows the RStudio interface with the following R code in the console:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[workspace loaded from ~/.RData]

> # Given data
> x <- c(21,24,27,29,31,34,37,41,45,48)
>
> # Construct Q-Q plot
> qqnorm(x)
> qqline(x, col = "red")
>
> # Comment
> cat("The points are approximately along the straight line.\n")
The points are approximately along the straight line.
> cat("Therefore, the data is approximately normally distributed.\n")
Therefore, the data is approximately normally distributed.
> |
```

The Environment pane on the right shows the following variables:

Variable	Class	Value
j	num	3
MAD	num	6.16326530612245
mean_x	num	24.1428571428571
normalized	num [1:5]	0.125 0.25 0.375 0.5 0.75
x	num [1:10]	21 24 27 29 31 34 37 41 45 48

The plot titled "Normal Q-Q Plot" shows Sample Quantiles on the y-axis (ranging from 20 to 45) and Theoretical Quantiles on the x-axis (ranging from -1.5 to 1.5). The data points are plotted as open circles, and a red line represents the theoretical normal distribution. The points closely follow the red line, indicating that the data is approximately normally distributed.

3. WEKA: Create patient Age, Glucose, BMI, Diabetes ARFF data; apply J48 and obtain confusion matrix.

Dataset: Diabetes

```
@relation diabetes
@attribute Age numeric
@attribute Glucose numeric
@attribute BMI numeric
@attribute Diabetes {Yes,No}
@data
25,95,22.5,No
32,130,28.1,Yes
41,145,31.2,Yes
29,105,24.8,No
50,160,34.5,Yes
36,120,27.0,No
45,150,32.1,Yes
27,100,23.5,No
55,170,36.0,Yes
39,115,26.5,No
```

The screenshot shows the Weka Explorer window with the 'Classify' tab selected. The classifier chosen is J48 with parameters -C 0.25 -M 2. The test options are set to 'Use training set'. The classifier output is displayed in the right pane, showing a size of the tree of 3 and a time taken to build the model of 0.01 seconds. The evaluation on the training set shows 100% accuracy. The detailed accuracy by class shows a TP Rate of 1.000, FP Rate of 0.000, Precision of 1.000, Recall of 1.000, F-Measure of 1.000, MCC of 1.000, ROC Area of 1.000, and PRC Area of 1.000 for both 'Yes' and 'No' classes. The confusion matrix shows 5 instances correctly classified as 'Yes' and 5 instances correctly classified as 'No'.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose J48 -C 0.25 -M 2

Test options

- ☒ Use training set
- ☐ Supplied test set
- ☐ Cross-validation Folds 10
- ☐ Percentage split % 66

More options...

(Nom) Diabetes

Start Stop

Result list (right-click for options)

- 13:24:27 - trees.J48

Classifier output

Size of the tree : 3

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	10	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Yes
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	No
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

```
a b <-- classified as
5 0 | a = Yes
0 5 | b = No
```

Status: OK

Log x 0

4. WEKA: Apply SMO to the diabetes data and calculate accuracy, precision, recall and F1; compare with J48.

Dataset: Diabetes – SMO

Use the same Diabetes ARFF dataset from Set 6 Q3.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseSMO -C 1.0 -L 0.001 -P 1.0E-12 -N 0 -V -1 -W 1 -K "weka.classifiers.functions.supportVector.PolyKernel -E 1.0 -C 250007" -calibrator "weka.classifiers.functions.Logistic -R 1.0E-

Test options

☒ Use training set

☐ Supplied test set

☐ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) Diabetes

Start

Stop

Result list (right-click for options)

13:24:27 - trees.J48

13:26:26 - functions.SMO

Classifier output

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	9	90	%
Incorrectly Classified Instances	1	10	%
Kappa statistic	0.8		
Mean absolute error	0.1		
Root mean squared error	0.3162		
Relative absolute error	20	%	
Root relative squared error	63.2456	%	
Total Number of Instances	10		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.800	0.000	1.000	0.800	0.889	0.816	0.900	0.900	Yes
	1.000	0.200	0.833	1.000	0.909	0.816	0.900	0.833	No
Weighted Avg.	0.900	0.100	0.917	0.900	0.899	0.816	0.900	0.867	

=== Confusion Matrix ===

a b <-- classified as

4 1 | a = Yes

0 5 | b = No

Status

OK

Log

x 0